

Recap

Regular expression $\longrightarrow$ Language
Created using $\{w \mid \text{DFA accepts } w\}$

- i Union : $\cup, +, |$
- ii Concatenation : $\circ$
- iii Kleene star : $*$

Today:

Language $\longrightarrow$ Regular expression

Operator precedence

- ① (R_1)
- ② R_1^*
- ③ $R_1 R_2$
- ④ $R_1 \cup R_2$

So $a.b^*.c \cup d$ is parsed as :

$$\left((a(b^*))c \right) \cup d$$

Regular expression

$$(0 \cup 1) = \Sigma$$

$$\{0\} \cup \{1\} = \{0, 1\} = \Sigma$$

The regular operations

Sipser Def 1.23 p. 44

For A, B languages over same alphabet, define:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\} = \text{Match pattern A or Match Pattern B}$$

$$A \circ B = \{xy \mid x \in A \text{ and } y \in B\} = \text{Match pattern A, then Match pattern B.}$$

Represents Sequencing

$$A^* = \{\underline{x_1 x_2 \dots x_k} \mid k \geq 0 \text{ and each } \textcircled{x_i} \in \textcircled{A}\}$$

Create Zero or More Copies of A.

$$A^3 = A \cdot A \cdot A \\ 010 = 010$$

$$\boxed{\{0, 1\}}$$

These are operations on sets!

Writing Regular Expression



Example

Write regular expression which generates

$$L_1 = \{w \mid \text{length of } w \text{ is 1 or more}\}$$

$$\Sigma = \{0, 1\}$$

The length of string should be ≥ 1

length one Regular expression : $(0 \cup 1) \rightarrow \{0, 1\}$
Re Language

$$(0 \cup 1)^* = \{0, 1\}^* = \Sigma^* = L^*$$

$$L^* = L^0 \cup L^1 \cup L^2 \cup L^3 \dots$$

$\downarrow$
 $\{\epsilon, 0, 1, 00, 01, 10, 11, \dots\}$

$\rightarrow$ length 3 strings

Final expression :

$$(0 \cup 1)(0 \cup 1)^* = \Sigma \Sigma^*$$

$\downarrow$
0 $\Sigma = 0$
1 $\Sigma = 1$

$$L^2 = L \cdot L$$
$$= \{0, 1\} \cdot \{0, 1\}$$
$$= \{00, 01, 10, 11\}$$

Writing Regular Expression



Example

Write regular expression which generates

$$\Sigma = \{0, 1\}$$

$L_2 = \{w \mid w \text{ contains at least two 1s}\}$

$$\geq 2 = \{2, 3, 4, \dots\}$$

$$L_2 = \{11, 01010, \dots\}$$

Any binary strings

1

Any binary strings

1

Any binary strings

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \dots$$

$$\underbrace{011}_{\Sigma^*}$$

1

101

1

11

$$\Sigma^2 = \Sigma \cdot \Sigma \cdot \Sigma$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$0 \quad 1 \quad 1$$

1

Σ^*

1

Σ^*

$$= \Sigma^* 1 \Sigma^* 1 \Sigma^*$$

input: 0 | 0 |

1

1

1

ϵ

Writing Regular Expression



Example $0^* = \{ \epsilon, 0, 00, 000, \dots \}$

$$\Sigma = \{0, 1\}$$

Write regular expression which generates

$$L_3 = \{w \mid w \text{ contains at most two 1s}\}$$

$\leq 2 \rightarrow \{0, 1, 2\}$

$$L_3 = \{ \epsilon, 1, 11, 010, \underline{001000100}, \dots \}$$

Contains 2 1's

RE : $0^* 1 0^* 1 0^*$

Diagram showing the structure of the regular expression $0^* 1 0^* 1 0^*$ with arrows indicating the sequence of characters and the possibility of multiple 0s. A red 'X' is marked below the expression.

Final expression:

$$0^*(1 \vee \epsilon) 0^* (1 \vee \epsilon) 0^*$$

$$0010 \checkmark$$
$$00000$$

Writing Regular Expression



If n is divided by d , there are d remainders

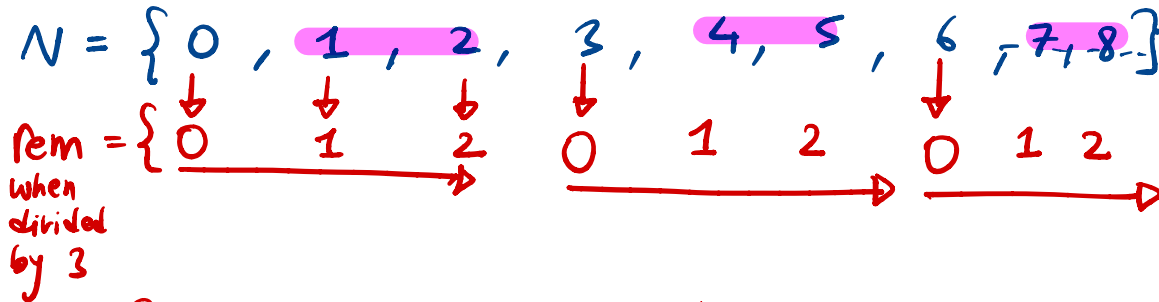
Example

$$0 \leq r < d$$

Write regular expression which generates

$$L_4 = \{w \mid \text{length of } w \text{ is not divisible by } 3\}$$

Remainder 0: The number is divisible by 3



$$0 \leq r < 3$$

0, 1, 2

quotient

$$\frac{12}{3} = \boxed{4} \text{ rem} = 0$$

$$n \rightarrow \frac{13}{3} = \boxed{4} 33 \dots \text{rem} = 1$$

Diagram illustrating the division of 13 by 3. The quotient is 4 and the remainder is 1. The remainder is calculated as $13 - 3 \times 4 = 1$.

Remainder 1 and 2 are not divisible by 3

$$\text{rem} = n - d \times q = 13 - 3(4) = 1$$

Writing Regular Expression

$$2^3 = 8$$



Example

Write regular expression which generates

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3$$

$L_4 = \{w \mid \text{length of } w \text{ is not divisible by } 3\}$

$$\downarrow$$
$$\Sigma . \Sigma . \Sigma$$

$$L = \{w \mid \text{length divisible by } 3\}$$

$$\boxed{\bar{L} = \Sigma^* - L} \times$$

$$(\underline{\Sigma . \Sigma . \Sigma})^* = \{ \underline{000, 001, 010, 011, 100, 101, 110, 111} \}^*$$

$$L^* = L^0 \cup \boxed{L^1} \cup L^2 \cup L^3 \dots$$

Σ length 3 length 6 length 9

Writing Regular Expression



Example

Write regular expression which generates

$$L_4 = \{w \mid \text{length of } w \text{ is not divisible by } 3\}$$

Write an expression for generating rem 1 and 2 lengths.

Rem 0	Rem 1	Rem 2
0	1	2
3	4	5
6	7	8
9	10	11

Match Rem 0 . length 1 = Rem 1

Match Rem 0 . length 2 = Rem 2

$$RE : (\Sigma\Sigma\Sigma)^* (\Sigma) \cup (\Sigma\Sigma\Sigma)^* (\Sigma\Sigma)$$

Writing Regular Expression

$+$, $\cup$, $|$ = OR

Example

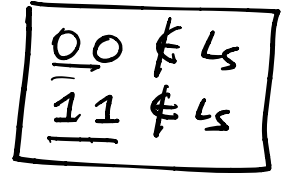
Write regular expression which generates

$$L_5 = \{w \mid 0s \text{ and } 1s \text{ alternate in } w\}$$

$$L_5 = \{01010, \dots, 1010\dots\}$$

$$(01)^*(0+\epsilon) + (10)^*(1+\epsilon)$$

$$\hookrightarrow \{\epsilon, 01, 0101, \dots\}$$



Writing Regular Expression



Example

Write regular expression which generates

$$\bar{L} = L_6 = \{w \mid w \text{ doesn't contain } 00 \text{ and } 11\}$$

$$L = \{w \mid w \text{ contains } 00 \text{ and } 11\}$$

$$R(L) = \Sigma^* 00 \Sigma^* 11 \Sigma^* + \Sigma^* 11 \Sigma^* 00 \Sigma^*$$

$$R(L_6) = (01)^*(0+\varepsilon) + (10)^*(1+\varepsilon)$$

Writing Regular Expression



Example

Write regular expression which generates

$$L_7 = \{w \mid w \text{ doesn't end with } \underline{01}\}$$

It can end with 00, 11, 10

$$\Sigma^* (00 + 11 + 10)^* + 0 + 1 + \epsilon$$

Input: 001011

1

Writing Regular Expression



Example

$11^* = 1^+ = \text{one or more } 1\text{'s}$
 $1^* = \text{zero or more } 1\text{'s}$

Write regular expression which generates

$$L_8 = \{w \mid w \text{ doesn't contain } \underline{00}\}$$

L_8 can contain $01, 10, 11$

$$R(L_8) = 1^*(01^*)^*(0+\varepsilon)$$

$10.01 \times$

$$(0+\varepsilon)11^*(0+\varepsilon)$$

0 | 0 | 111 | 0 | alternating 0 and 1 missing, any amount of 1's
can be there.

Writing Regular Expression



Example

Write regular expression which generates

$$L_9 = \{w \mid w \text{ doesn't contain } 10\}$$

L_9 can contain 00, 11, 01

0* 1*

0* 1*

000 111

00000 1